SI Appendix for
Hopkins, Melanie J, and Smith, Andrew B. Dynamic evolutionary change in post-Paleozoic echinoids and the importance of scale when interpreting changes in rates of evolution.

Corrections to character matrix
Before running any analyses, we corrected a few errors in the published character matrix of Kroh and Smith (1). Specifically, we removed the three duplicate records of *Oligopygus*, *Haimea*, and *Conoclypus*, and removed characters C51 and C59, which had been excluded from the phylogenetic analysis but mistakenly remain in the matrix that was published in Appendix 2 of (1). We also excluded *Anisocidaris*, *Paurocidaris*, *Pseudocidaris*, *Glyphopneustes*, *Enichaster*, and *Tiarechinus* from the character matrix because these taxa were excluded from the strict consensus tree (1). This left 164 taxa and 303 characters for calculations of rates of evolution and for the principal coordinates analysis.

Other tree scaling methods
The most basic method for scaling a tree using first appearances of taxa is to make each internal node the age of its oldest descendent ("stand") (2), but this often results in many zero-length branches which are both theoretically questionable and in some cases methodologically problematic (3). Several methods exist for modifying zero-length branches. In the case of the results shown in Figure 1, we assigned a positive length to each zero-length branch by having it share time equally with a preceding, non-zero-length branch (“equal”) (4). However, we compared the results from this method of scaling to several other methods. First, we compared this with rates estimated from trees scaled such that zero-length branches share time proportionally to the amount of character change along the branches (“prop”) (5), a variation which gave almost identical results as the method used for the “equal” method (Fig. S3E-F). A fundamentally different way of treating zero-length branches is to assign an arbitrarily small positive length to zero-length branches. This value may be added to just the zero-length branches (“zbla”), added equally to all branches in the tree (“aba”), or applied by scaling all branches so that they are greater than or equal to a small positive length while subtracting time added to later branches from earlier branches in order to maintain the temporal structure of events (“mbl”) (3, 6). Finally, Bapst recently developed a stochastic time-scaling method where the selection of node ages is weighted by a probability density function derived from branching, extinction, and sampling rates estimated from the taxon ranges (cal3, 7, 8). We implemented this method in the following way. First, we assigned numeric ages to the first appearances, as well as the last appearances, as described in the Methods, and estimated sampling and extinction rates from the distribution of extinct taxon durations (9); branching rates were set to be equal to extinction rates. Then we scaled each tree under two different variants of the cal3 method, one that considers potential ancestral relationships (allowing for budding cladogenesis *sensu* Foote [10], and referred to as “cal3-withA”) and one that does not (“cal3-noA”). Because all character changes along a branch leading up to a taxon have taken place by the time of the first appearance of that taxon, times of observation were constrained to the first appearances. Rates of character change estimated from differently-scaled trees are shown in Figure S3.

We did not use Bayesian methods that simultaneously infer topology and divergences times (e.g., ref 11) in order to produce time-scaled trees because a well-supported and well-resolved strict consensus tree generally consistent with both the fossil record and analyses based on molecular data was already available (1).

Time series analysis

In order to assess whether changes in rates of character change between time intervals was influenced by sampling completeness, we compared the time series of rate changes to two measures of sampling variation: 1) the number of collections that included echinoid taxa based on records downloaded from the Paleobiology Database (www.paleobiodb.org) on 12 November 2014 (Table S4); and 2) the number of lineages sampled within each time interval divided by the number of lineages inferred to be present from the phylogenetic analysis (Table S3). The former we refer to as "sampling intensity" and the latter as "completeness". In order to stabilize variance, we power-transformed rates of character change and sampling intensity using the Box-Cox transformation, and logit-transformed completeness (a proportion), remapping logit transformations to the interval 0.025-0.975 in order to retain estimates of 1 after the transformation (12). We then calculated Spearman's rank correlations between generalized differences (13) of the two sets of time series. Time series and bivariate plots are shown in Figure S6; correlations are statistically insignificant regardless of tree-scaling method (Table S5). We also computed cross-correlations in order to determine if there were any significant lagged relationships between the rates and the different sampling metrics.

Branch likelihood test

The likelihood-based approach adopted here is described in detail in Lloyd *et al.* (14). Briefly, for any given branch, the number of character changes occurring along that branch is modeled as a Poisson process with rate parameter λ. λ is the expected number of character changes per lineage million years if all of the characters were observable. The expected number of changes per branch is then the product of the expected rate (λ), the duration of the branch in millions of years, and the proportion of the total number of characters observed on that branch. The hypothesis that any particular branch significantly differs in its rate from the rest of the tree is tested using a likelihood ratio where the numerator is the likelihood evaluated if all of the branch rates are the same (the null hypothesis) and the denominator is the likelihood evaluated if all of the branch rates are the same except for the branch of interest (the alternative hypothesis). The test statistic follows a chi-square distribution with one degree of freedom under the null hypothesis. Because there are multiple comparisons, the false discovery rate is controlled for following Benjamini and Hochberg (15).

Gower's coefficient and principal coordinates analysis

For taxa *i* and *j* with *v* characters *k*, Gower's coefficient is:

$$S_{ij} = \sum_{k=1}^{v} S_{ijk} \Big/ \sum_{k=1}^{v} \delta_{ijk}$$

Typically S_{ijk} = 1 if the kth characters agree and 0 otherwise, and δ_{ijk} = 1 when the characters are comparable and 0 when they are not (for example, the character state for one taxon is missing). For this analysis, we were interested in dissimilarity between taxa, so if either character was scored as missing, unknown, polymorphic, or uncertain, or the characters do not match, S_{ijk} = 1; if both kth characters match exactly, S_{ijk} = 0. Because polymorphisms and uncertain character states are relatively uncommon (where they occur, they account for on average 1% and 0.7% of character states within any given taxon, respectively, and each account for about 0.1% percent of the character states across the entire matrix), treating them in a different way would have had a

negligible effect on the results. Gower's coefficient was calculated using the daisy function in the cluster package for R (16).

Use of Gower's coefficient as a dissimilarity metric may result in non-Euclidean dissimilarities. If the dissimilarities are Euclidean, the distances between points in the ordination will be exactly equal to the dissimilarities between the taxa. This equality becomes approximate if the dissimilarities are non-Euclidean, and ordinating such a matrix will result in negative eigenvalues. The effects are minimal as long as the negative eigenvalues are small in magnitude (17, 18), and the number of negative eigenvalues can be reduced by adding a constant to the non-diagonal dissimilarities such that the modified dissimilarities are Euclidean (18, 19). Although our dissimilarity matrix produced negative eigenvalues, their magnitudes were small, and we found that adding a constant (computed analytically following Cailliez [19] and implemented in cmdscore function in R) had a negligible effect on the principal coordinates scores and no effect on the disparity calculations.

References

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Figure and Table Captions

Figure S1. Strict consensus of tree (1) scaled using first appearances of genera and treating zero-length branches using the “equal” method (4). Note that there are only a few local polytomies.

Figure S2. Comparison of different protocol for estimating rates of morphological evolution. (**A**) Rates of character change in echinoids based on 1000 iterations for each tree (100000 time series in total). (**B**) Rates of character change in echinoids, computed from character changes mapped onto the phylogeny using DELTRAN optimization. (**C**) Rates of character change in echinoids, computed from a tree that was scaled using family-level first appearance data. (**D**) Rates of character change in echinoids, computed for a high-resolution timescale (time intervals are approximately 5 million years long, Table S7). Error bars represent 95% confidence intervals. Black points = mean rate values; blue points = median rate values. Compare with Figure 1 which is based on the same data but is based on only 100 iterations for each tree, characters were mapped using ACCTRAN optimization, the tree was scaled using genus-level first appearance data, and time intervals are approximately 10 million years long.

Figure S3. Rates of morphological evolution in echinoids since the Paleozoic, computed from a tree where (**A**) zero-length branches were retained without adjustment ("stand") (2); (**B**) where zero-length branches were adjusted by adding an arbitrary constant minimum value to zero-length branches only (“zlba”) (20); (**C**) zero-length branches were adjusted by adding an arbitrary constant minimum value to all branches (“aba”) (20); (**D**) zero-length branches were adjusted by scaling all branches so that they are greater or equal to an arbitrary minimum while subtracting time added to later branches from earlier branches in order to maintain the temporal structure of events (“mbl”) (6); (**E**) zero-length branches were adjusted by sharing time equally with a preceding, non-zero-length branch (“equal”) (4); (**F**) zero-length branches were adjusted by sharing time with a preceding, non-zero duration branch in proportion to the amount of character change occurring along both branches (“prop”) (5); (**G**) zero-length branches were modified using the cal3 method excluding ancestors (“cal3-noA”) (7); and (**H**) zero-length branches were modified using the cal3 method including ancestors (“cal3-with A”) (7). Black points = mean rate values; blue points = represent median rate values, plotted in order to demonstrate that most rate estimates in the Triassic are low despite high confidence intervals using some scaling methods (compare with **E**, which is the same as Figure 1 but rescaled for easy comparison to other methods).

Figure S4. Strict consensus trees pruned to include only living taxa. (**A**) Taxa pruned after scaling the tree to taxon first appearances and treating zero-length branches using the “equal” method (4). (**B**) Taxa pruned to include only living taxa and then tree scaled to their first appearances as observed in the fossil record and treating zero-length branches using the “equal” method (4). Thick black lines indicate stratigraphic ranges but tips were all dated at 0 when estimating rates through time.

Figure S5. Rates of evolution in echinoids since the Paleozoic, computed from a tree pruned to just living taxa and then scaled using the first appearances of those taxa as inferred from the fossil record (**A**) and computed from a tree pruned to just living taxa but after the scaling of the complete tree (**B**). Compare with Figure 1, which was computed using a tree including fossil and living taxa.

Figure S6. Results of time series analyses. (**A**) Plot of time series: rates of character change based on means shown in Figure 1; completeness of the echinoid fossil record measured as the ratio of sampled lineages to the number of lineages inferred to be present based on phylogenetic analysis; sampling intensity based on the number of collections including echinoids made during each time interval. (**B**) Bivariate plot of generalized first differences of rates of character change and completeness. (**C**) Bivariate plot of generalized first differences of rates of character change and sampling intensity.

Figure S7. Differences in rates of morphological evolution in different echinoid groups since the Paleozoic: (**A**) All regular echinoids; (**B**) Non-clypeasteroid irregular echinoids; (**C**) Atelostomata (heart urchins and their relatives); and (**D**) Neognathostomata (clypeasteroids and their relatives). See Figure 2 for placement of groups on phylogenetic tree and rates of character change in additional groups.

Figure S8. Taxonomic diversification of echinoids through time, estimated by tabulating the number of first appearances of genera in each time interval (timescale shown in Table S2). Data from Kroh and Smith (1, appendix 3). Dashed line shows increase to the present, which is inflated because it includes genera with no known fossil record (and therefore could have evolved earlier than the last 10 million years).

Figure S9. Phylomorphospace based on principal coordinates analysis of character matrix, showing morphological disparity of post-Paleozoic echinoids during different geological periods. The first two principal coordinates together summarize 57.3% of the variation. Warm colors = irregular echinoids: red = Neognathostomata; orange = Atelostomata; pink = stem groups. Cool colors = regular echinoids: cyan = Cidaroida; blue = regular euechinoids. See Figure 3 for morphospace occupation as realized across the entire clade and Figure 4 for group disparity estimates.

Table S1. First appearances of genera and families represented in the phylogenetic analysis. Ma = millions of years ago; L = Lower; M = Middle; U = Upper. Lower and upper bounds based on Gradstein *et al.* (21).

Table S2. Stage-based timescale where average time interval is about 10 million years. Mya = millions of years ago; my = millions of years. Lower and upper bounds based on Gradstein *et al.* (21).

Table S3. Computation of phylogeny-based measure of completeness. 1 = species within clade recorded from that time interval; 0 = no species within clade recorded from that time interval. Lower and upper bounds based on Gradstein *et al.* (21).

Table S4. Numbers of collections yielding specific echinoid clades. Tabulation based on records downloaded from the Paleobiology Database (www.paleobiodb.org) on 12 November 2014. Lower and upper age boundaries based on Gradstein *et al.* (21).

Table S5. Correlations between rates of character change and sampling intensity (upper panel) and rates of character change and completeness (lower panel). Shorthand for scaling methods is followed by panel label in Figure S3.

Table S6. Tabulation of character changes during the early evolution of clypeasteroids. Below are listed all characters associated with the first 7 dichotomies at the base of the clypeasteroids (from the oligopygoid-clypeasteroid divergence to the establishment of the three major subclades, Clypeasterina, Laganiformes, and Scutteliformes). There are 51 characters in all. We scored each as (i) feeding related (e.g., peristome shape, tube-foot arrangement, lantern or perignathic girdle features); (ii) hydrodynamic features (i.e. to do with life in high energy environments, e.g., test flattening, internal butressing, lunules); (iii) neutral with respect to (i) and (ii) (e.g., detailed plating differences that have no clear functional correlate). Characters A27 and A28 are conservatively scored as neutral although it is possible that they are related to ciliary current generation during feeding.

Table S7. Stage-based timescale where average time interval is about 5 million years. Mya = millions of years ago; my = millions of years. Lower and upper bounds based on Gradstein *et al.* (21).

Figure S1

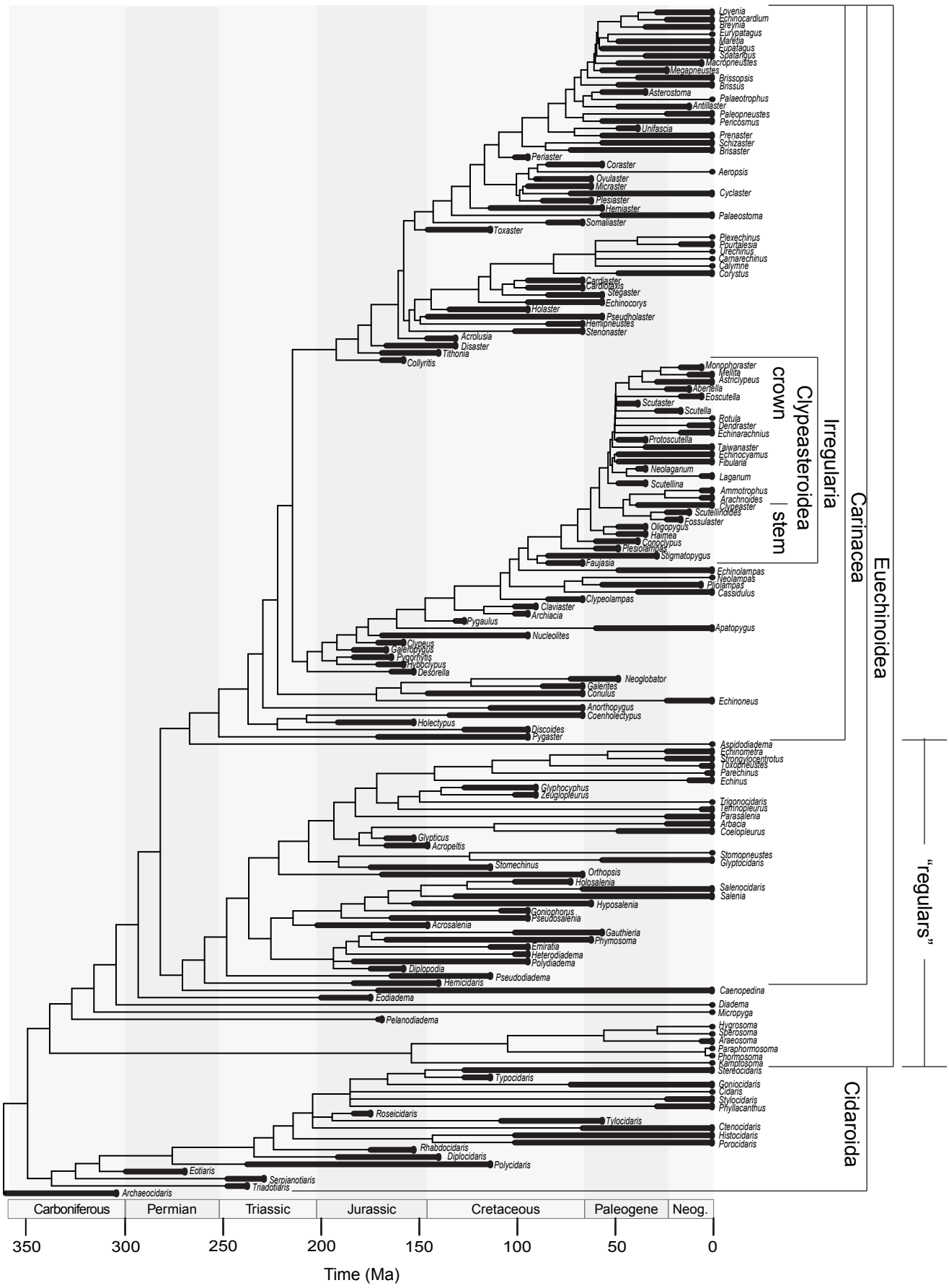

Lovenia
Echinocardium
Breynia
Eurypatagus
Maretia
Eupatagus
Spatangus
Necropneustes
Megapneustes
Brissopsis
Brissus
Asterostoma
Palaeotrophus
Antillaster
Paleopneustes
Pericosmus
Unifascia
Prenaster
Schizaster
Brisaster
Periaster
Coraster
Aeropsis
Ovulaster
Micraster
Cyclaster
Plesiaster
Hemiaster
Palaeostoma
Somaliaster
Toxaster
Plexechinus
Courtalesia
Urechinus
Camarechinus
Calymne
Corystus
Cardiaster
Cardiolaxis
Stegaster
Echinocorys
Holaster
Pseudholaster
Hemipneustes
Stenonaster
Acrolusia
Disaster
Tithonia
Collyritis
Monophoraster
Mellita
Astriclypeus
Abertella
Eoscutella
Scutaster
Scutella
Rotula
Dendraster
Echinarachnius
Protoscutella
Taiwanaster
Echinocyamus
Fibularia
Neolaganum
Laganum
Scutellina
Ammotrophus
Arachnoides
Clypeaster
Scutellinoides
Fossulaster
Oligopygus
Haimea
Conoclypus
Plesiolampas
Stigmatopygus
Faujasia
Echinolampas
Neolampas
Ptolampas
Cassidulus
Clypeolampas
Claviaster
Archiacia
Pygaulus
Apatopygus
Nucleolites
Clypeus
Galeropygus
Pygorhytis
Phrygoclypus
Desorella
Neoglobator
Galerites
Conulus
Echinoneus
Anorthopygus
Coenholectypus
Holectypus
Discoides
Pygaster
Aspidodiadema
Echinometra
Strongylocentrotus
Toxopneustes
Parechinus
Echinus
Glyphocyphus
Zeuglopleurus
Trigonocidaris
Temnopleurus
Parasalenia
Arbacia
Coelopleurus
Glypticus
Acropeltis
Stomopneustes
Glyptocidaris
Stomechinus
Orthopsis
Holosalenia
Salenocidaris
Salenia
Hyposalenia
Goniophorus
Pseudosalenia
Acrosalenia
Gauthieria
Phymosoma
Emiratia
Heterodiadema
Polydiadema
Diplopodia
Pseudodiadema
Hemicidaris
Caenopedina
Eodiadema
Diadema
Micropyga
Pelanodiadema
Hygrosoma
Sperosoma
Araeosoma
Paraphormosoma
Phormosoma
Kamptosoma
Stereocidaris
Typocidaris
Goniocidaris
Cidaris
Stylocidaris
Phyllacanthus
Roseicidaris
Tylocidaris
Ctenocidaris
Histocidaris
Porocidaris
Rhabdocidaris
Diplocidaris
Polycidaris
Eotiaris
Serpianotiaris
Triadotiaris
Archaeocidaris
Clypeasteroidea crown
stem
Irregularia
Carinacea
Euechinoidea
"regulars"
Cidaroida
Carboniferous
Permian
Triassic
Jurassic
Cretaceous
Paleogene
Neog.
350
300
250
200
150
100
50
0
Time (Ma)

Figure S2

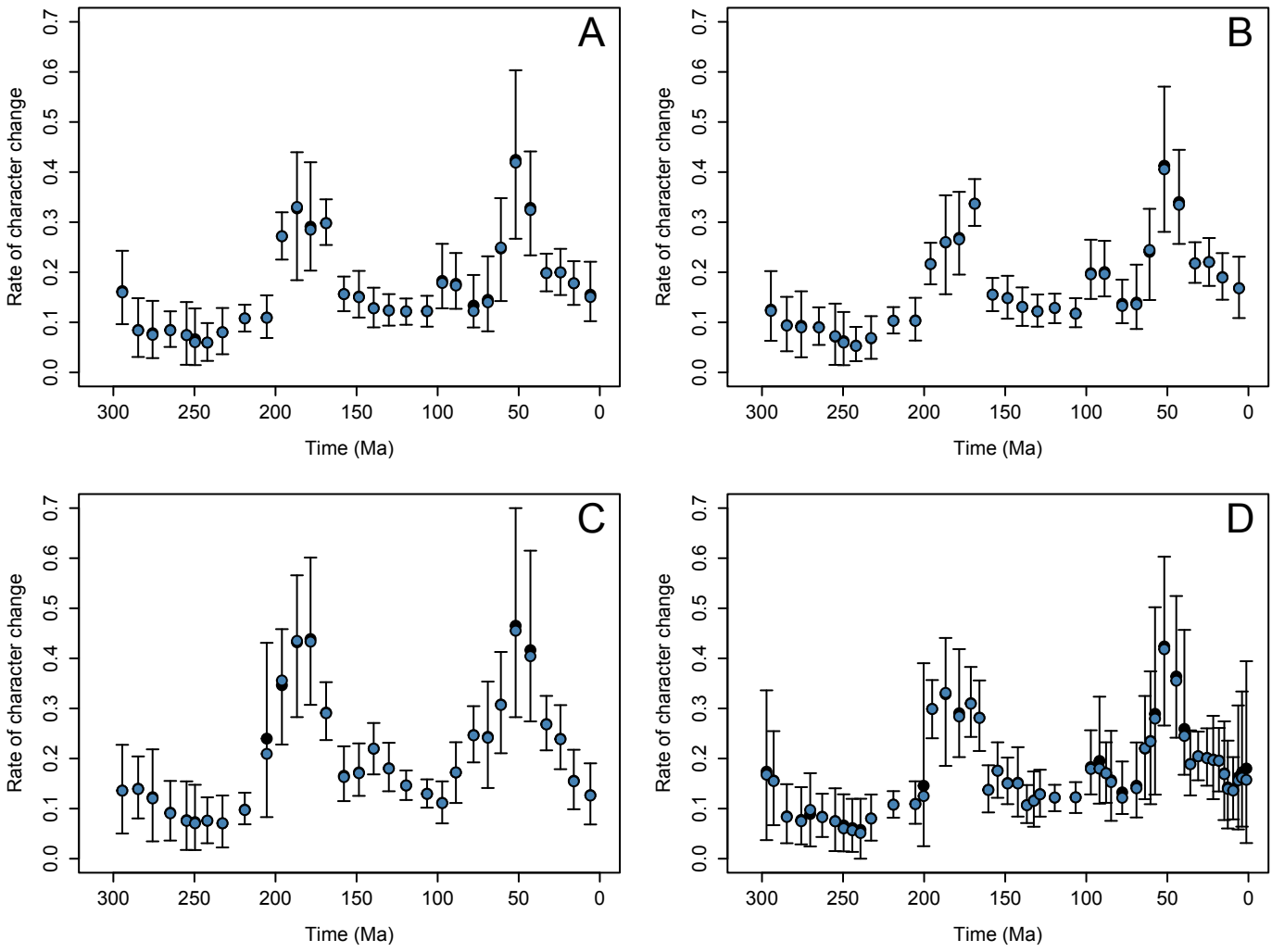

A
B
C
D
Rate of character change
Time (Ma)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
300
250
200
150
100
50
0

Figure S3

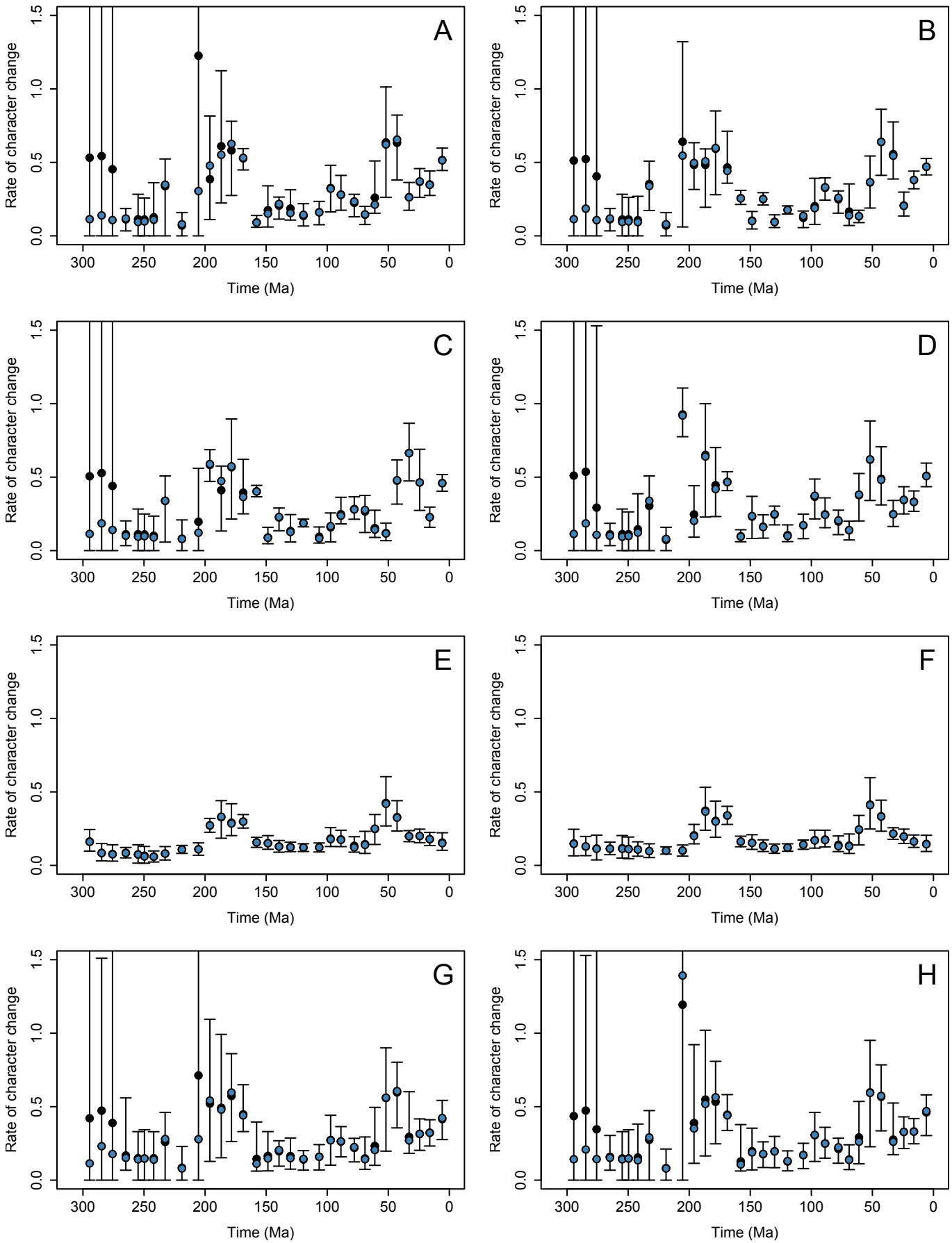

A
B
C
D
E
F
G
H
Rate of character change
Time (Ma)
0.0
0.5
1.0
1.5
300
250
200
150
100
50
0

Figure S4

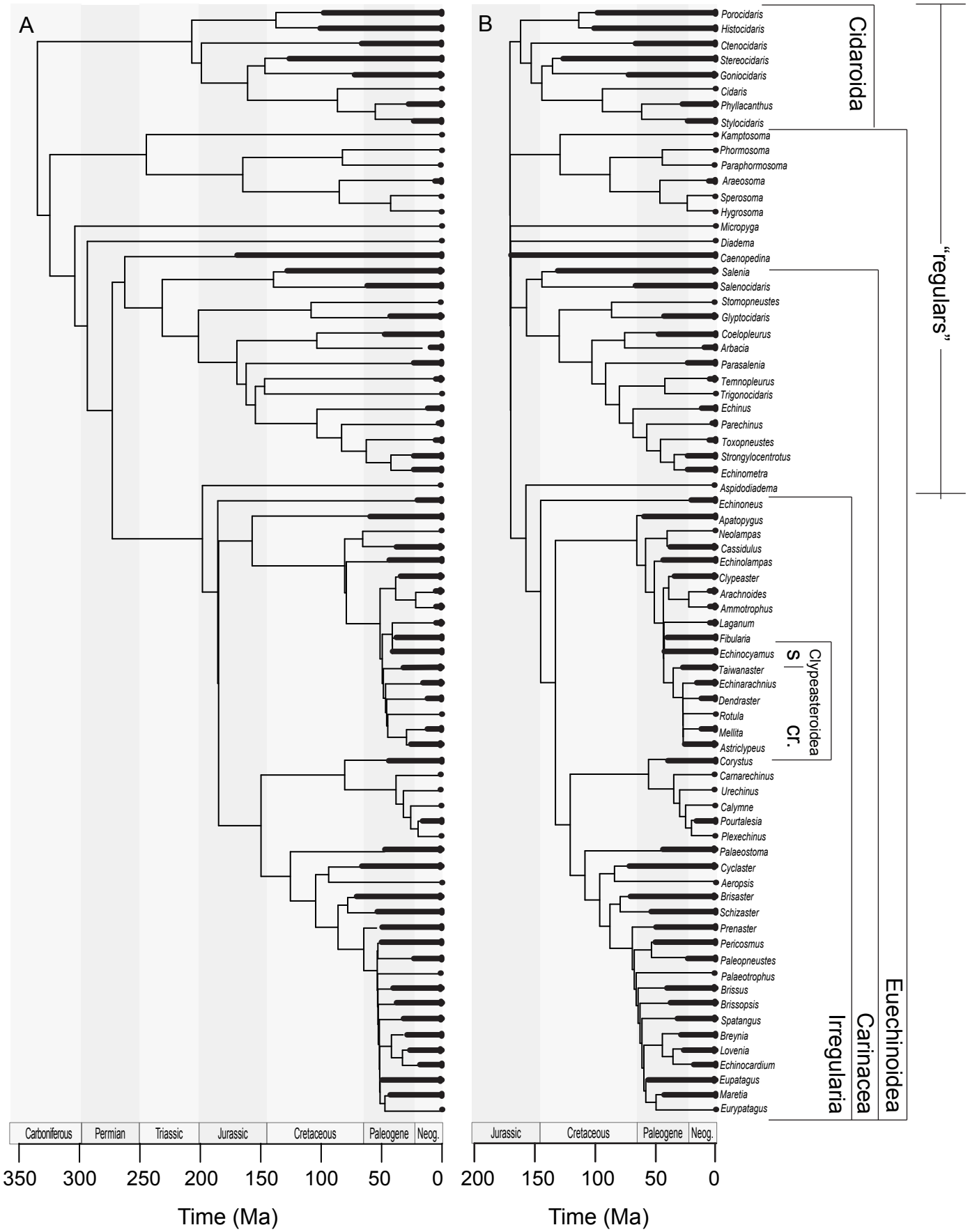

A
B
Porocidaris
Histocidaris
Ctenocidaris
Stereocidaris
Goniocidaris
Cidaris
Phyllacanthus
Stylocidaris
Kamptosoma
Phormosoma
Paraphormosoma
Araeosoma
Sperosoma
Hygrosoma
Micropyga
Diadema
Caenopedina
Salenia
Salenocidaris
Stomopneustes
Glyptocidaris
Coelopleurus
Arbacia
Parasalenia
Temnopleurus
Trigonocidaris
Echinus
Parechinus
Toxopneustes
Strongylocentrotus
Echinometra
Aspidodiadema
Echinoneus
Apatopygus
Neolampas
Cassidulus
Echinolampas
Clypeaster
Arachnoides
Ammotrophus
Laganum
Fibularia
Echinocyamus
Taiwanaster
Echinarachnius
Dendraster
Rotula
Mellita
Astriclypeus
Corystus
Carnarechinus
Urechinus
Calymne
Pourtalesia
Plexechinus
Palaeostoma
Cyclaster
Aeropsis
Brisaster
Schizaster
Prenaster
Pericosmus
Paleopneustes
Palaeotrophus
Brissus
Brissopsis
Spatangus
Breynia
Lovenia
Echinocardium
Eupatagus
Maretia
Eurypatagus
Cidaroida
"regulars"
Euechinoidea
Carinacea
Irregularia
Clypeasteroidea s. cr.
Carboniferous
Permian
Triassic
Jurassic
Cretaceous
Paleogene
Neog.
350
300
250
200
150
100
50
0
Time (Ma)
Jurassic
Cretaceous
Paleogene
Neog.
200
150
100
50
0
Time (Ma)

Figure S5

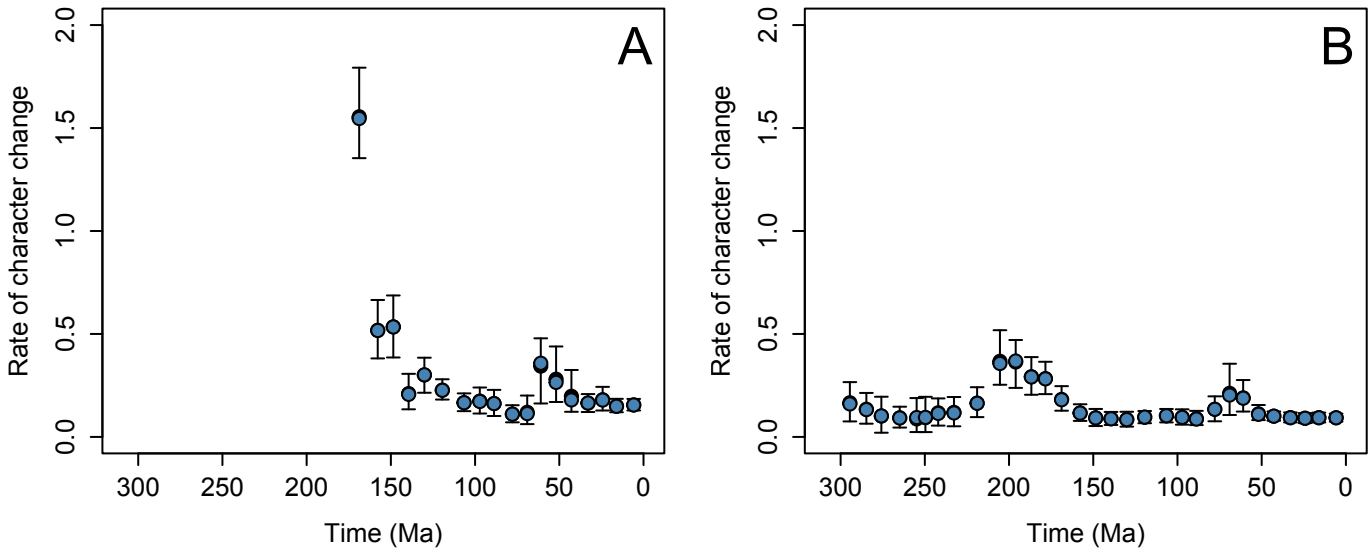

A
B
Rate of character change
0.0
0.5
1.0
1.5
2.0
300
250
200
150
100
50
0
Time (Ma)

Figure S6

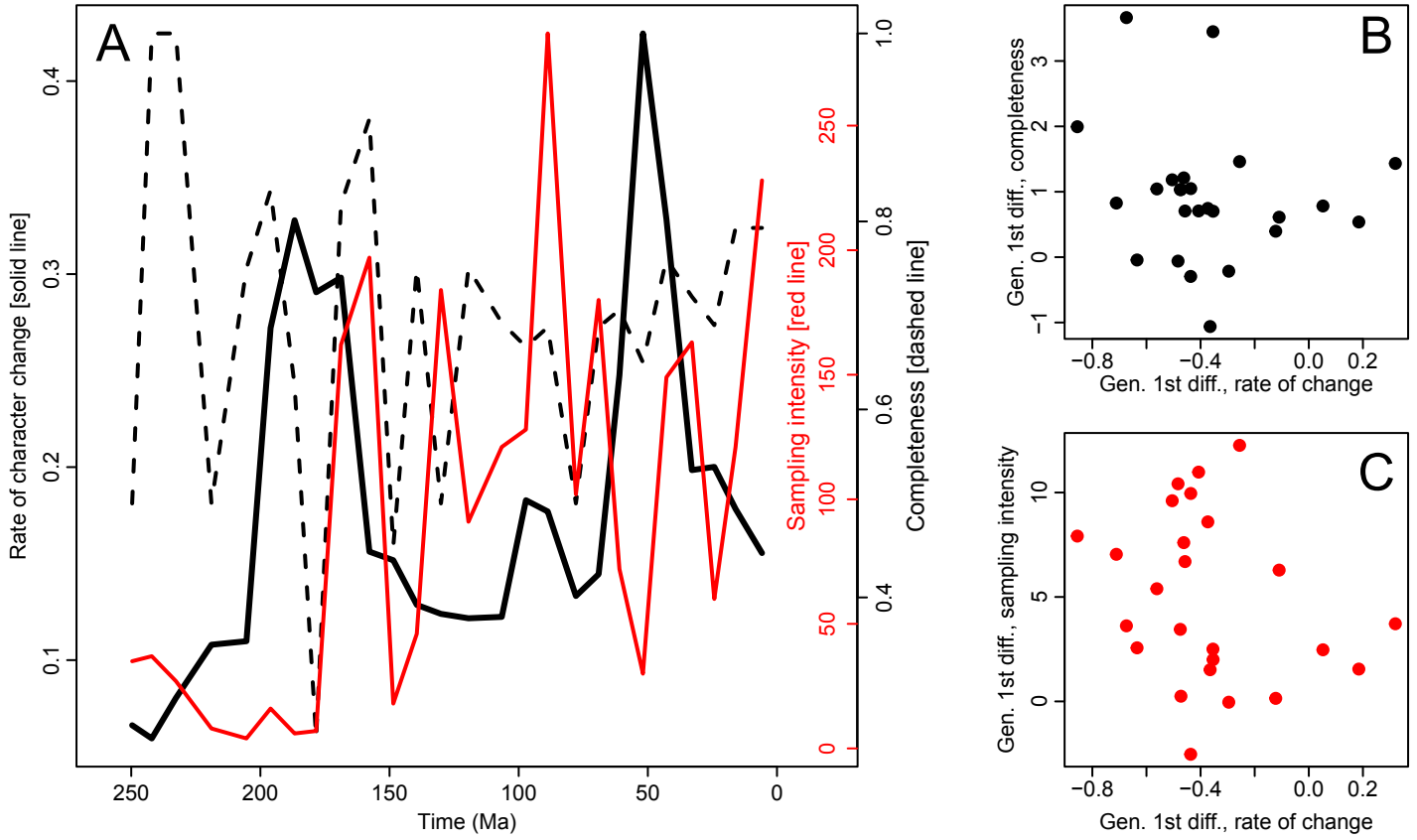

A
Rate of character change [solid line]
Sampling intensity [red line]
Completeness [dashed line]
Time (Ma)
B
Gen. 1st diff., completeness
Gen. 1st diff., rate of change
C
Gen. 1st diff., sampling intensity
Gen. 1st diff., rate of change

Figure S7

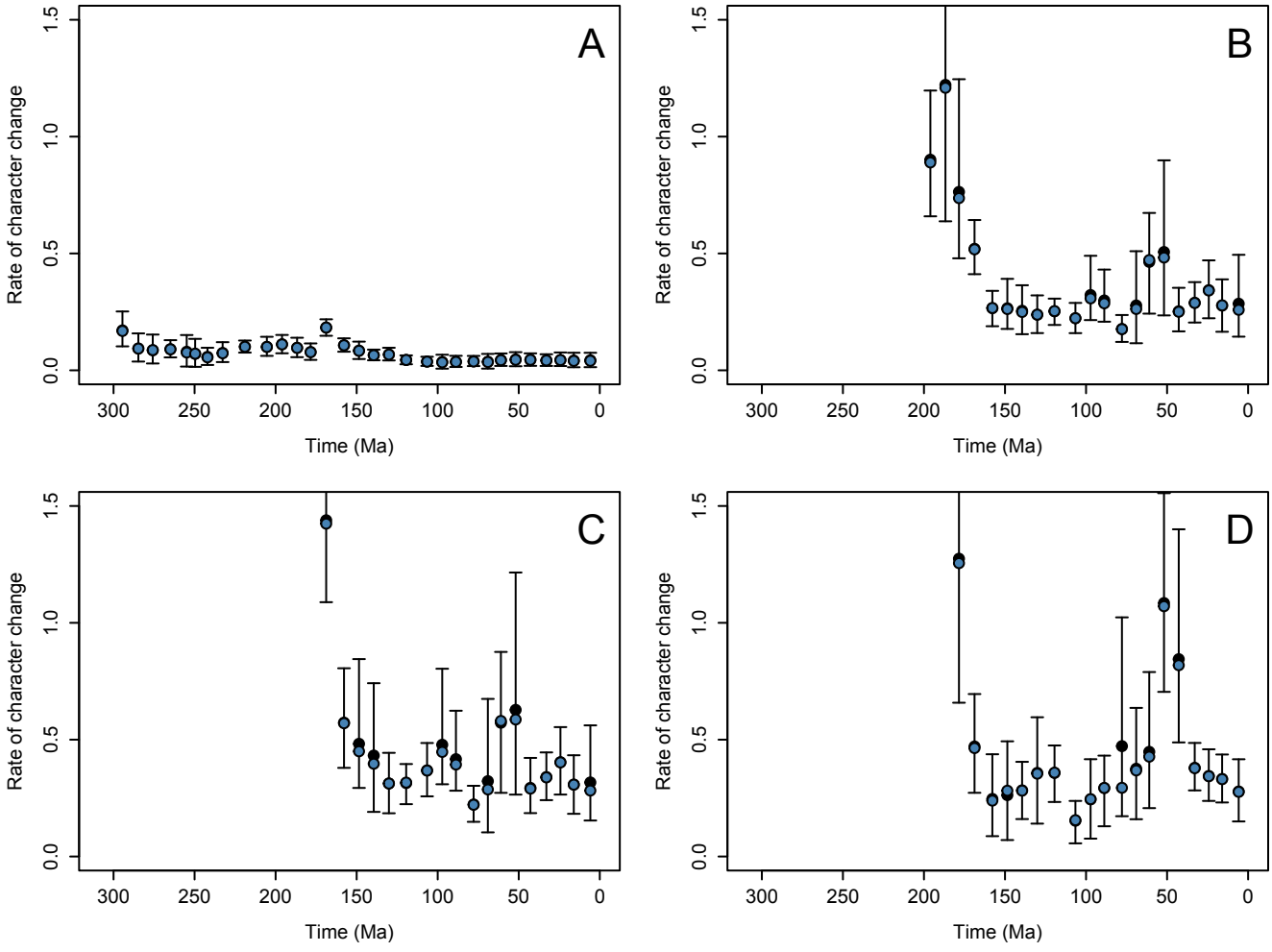

A
B
C
D
Rate of character change
Time (Ma)
0.0
0.5
1.0
1.5
300
250
200
150
100
50
0

Figure S8

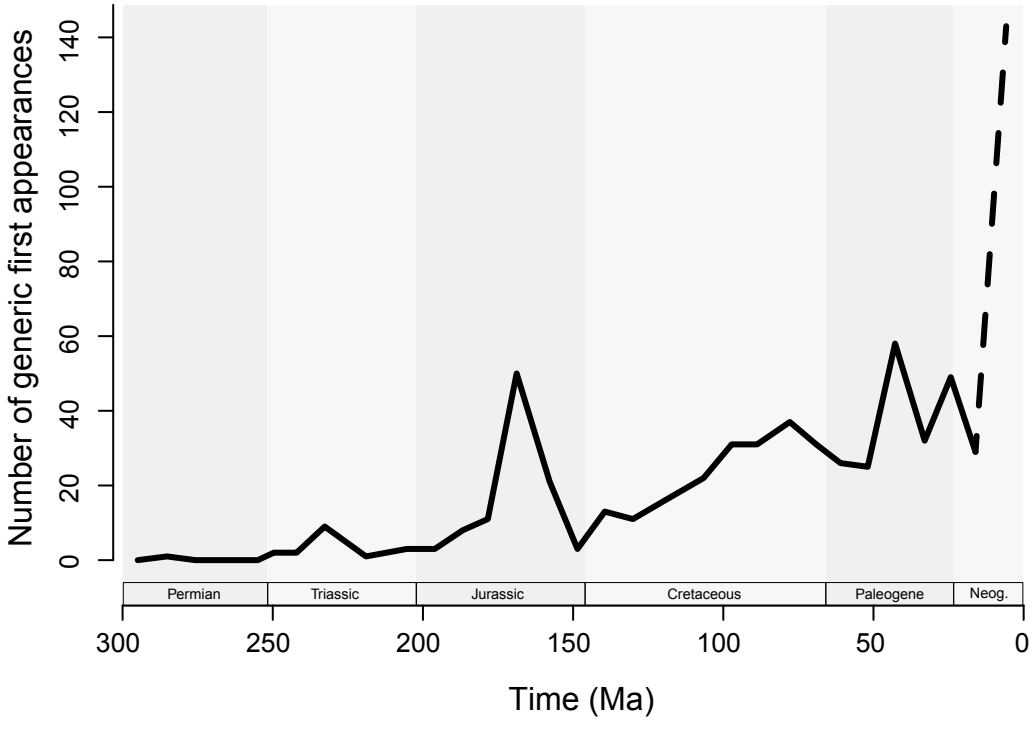

Number of generic first appearances
0
20
40
60
80
100
120
140
Permian
Triassic
Jurassic
Cretaceous
Paleogene
Neog.
300
250
200
150
100
50
0
Time (Ma)

Figure S9

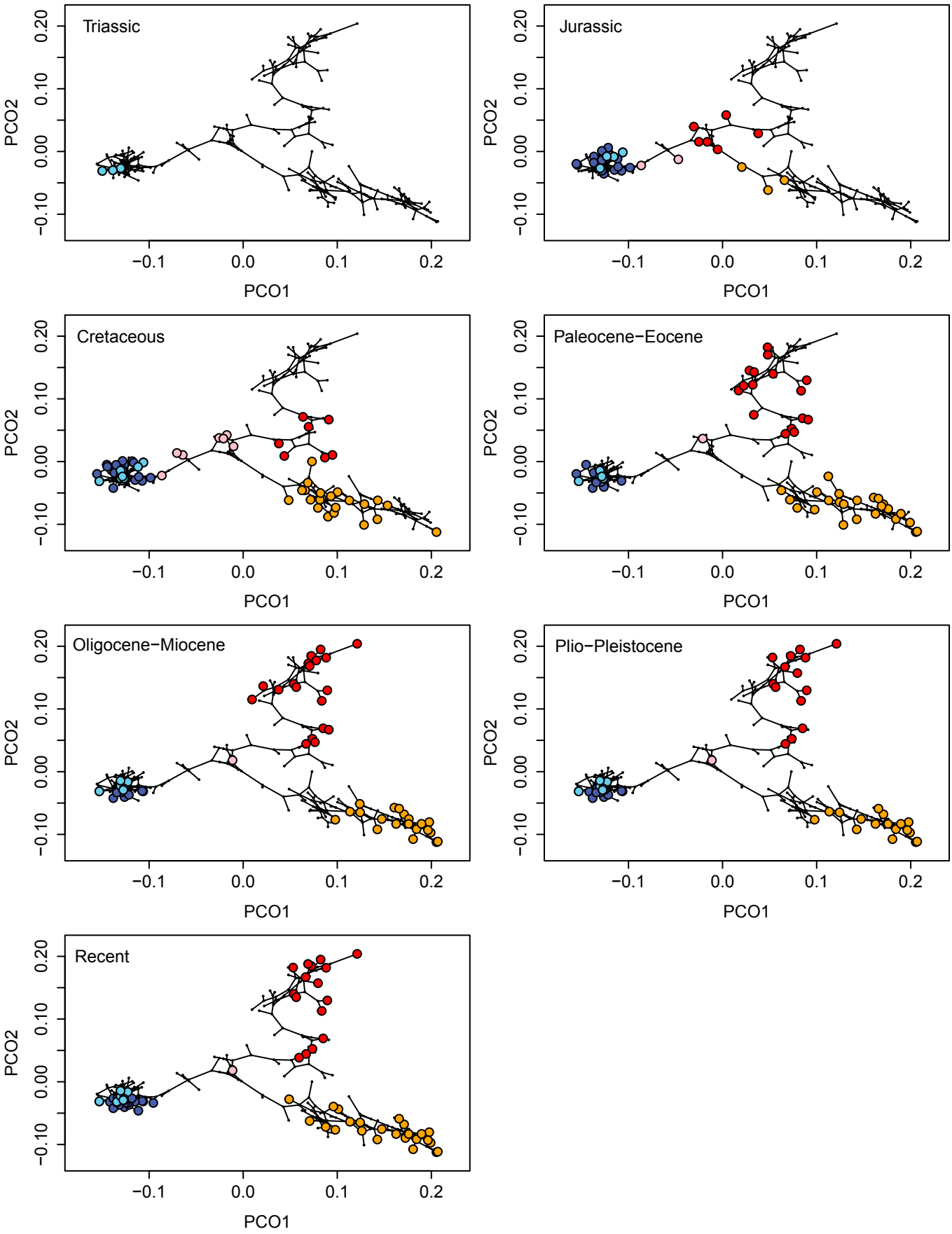

Triassic
Jurassic
Cretaceous
Paleocene−Eocene
Oligocene−Miocene
Plio−Pleistocene
Recent
PCO1
PCO2
−0.1
0.0
0.1
0.2
−0.10
0.00
0.10
0.20

Table S1. First appearances of genera represented in phylogenetic analysis and their respective families. Ma= millions of years ago; L = Lower; M = Middle; U = Upper. Lower and upper bounds based on Gradstein et al (21).

Genus	First appearance of genus in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)	Family	First appearance of family in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)
Abertella	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Abertellidae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Acrolusia	Neocomian = Berriasian-Hauterivian	145.3	130.8	Acrolusiidae	Neocomian = Berriasian-Hauterivian	145	130.8
Acropeltis	Callovian	166.1	163.5	Acropeltinae	Callovian	166.1	163.5
Acrosalenia	Hettangian	201.3	199.3	Acrosaleniidae	Hettangian	133.9	130.8
Aeropsis	Recent	0	0	Aeropsidae [Aeropidae]	Paleocene	66	23.03
Ammotrophus	Pliocene	5.333	2.588	Ammotrophinae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Anorthopygus	Albian	113	100.5	Anorthopygidae	Oxfordian	163.5	157.3
Antillaster	M Eocene = Lutetian-Bartonian	47.8	37.8	Antillasterinae	M Eocene = Lutetian-Bartonian	47.8	37.8
Apatopygus	U Paleocene = Thanetian	59.2	56	Apatopygidae	Paleocene	66	23.03
Arachnoides	Pliocene	5.333	2.588	Arachnoididae	Pliocene	5.333	2.588
Araeosoma	Pliocene	5.333	2.588	Echinothuriidae	M Miocene = Langhian-Serravalian	15.97	11.63
Arbacia	Miocene	23.03	5.333	Arbaciidae	Barremian	130.8	126.3
Archaeocidaris	Tournasian	358.9	346.7	Archaeocidaridae	Tournasian	358.9	346.7
Archiacia	Cenomanian	100.5	93.9	Archiaciidae	Cenomanian	100.5	93.9
Aspidodiadema	Recent	0	0	Aspidodiadematidae	Pliensbachian	190.8	182.7
Asterostoma	Eocene	56	33.9	Asterostomatidae	Eocene	56	33.9
Astriclypeus	U Oligocene = Chattian	28.1	23	Astriclypeidae	U Oligocene = Chattian	28.1	23
Breynia	Oligocene	33.9	23.03	Loveniidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Brisaster	Maastrichtian	72.1	66	[Brisasterinae]	Maastrichtian	72.1	66
Brissopsis	U Eocene = Priabonian	37.8	33.9	Brissopsidae	U Eocene = Priabonian	37.8	33.9
Brissus	M Eocene = Lutetian-Bartonian	47.8	37.8	Brissidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Caenopedina	Bajocian	170.3	168.3	Pedinidae	Rhaetian	209.5	201.3
Calymne	Recent	0	0	Calymnidae	Campanian	83.6	72.1
Cardiaster	Turonian	93.9	89.8	Cardiasteridae	Cenomanian	100.5	93.9
Cardiotaxis	Turonian	93.9	89.8	[Cardiotaxinae]	Turonian	93.9	89.8
Carnarechinus	Recent	0	0	Urechinidae	Recent	0	0
Cassidulus	U Eocene = Priabonian	37.8	33.9	Cassidulidae	Maastrichtian	72.1	66
Cidaris	Recent	0	0	Cidaridae	Bajocian	170.3	168.3
Claviaster	Cenomanian	100.5	93.9	Archiaciidae	Cenomanian	100.5	93.9

Genus	First appearance of genus in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)	Family	First appearance of family in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)
Clypeaster	U Eocene = Priabonian	37.8	33.9	Clypeasteridae	U Eocene = Priabonian	37.8	33.9
Clypeolampas	Campanian	83.6	72.1	Clypeolampadidae	Coniacian	89.8	86.3
Clypeus	Bajocian	170.3	168.3	Clypeidae	Bajocian	170.3	168.3
Coelopleurus	M Eocene = Lutetian-Bartonian	47.8	37.8	Arbaciidae	Barremian	130.8	126.3
Coenholectypus	Hauterivian	133.9	130.8	Coenholectypidae	Hauterivian	133.9	130.8
Collyrites	Bathonian	168.3	166.1	Collyritidae	Bathonian	168.3	166.1
Conoclypus	U Paleocene = Thanetian	59.2	56	Conoclypidae	Paleocene	66	23.03
Conulus	Valanginian	145	139.4	Conulidae	Oxfordian	163.5	157.3
Coraster	Campanian	83.6	72.1	Corasterinae	Coniacian	89.8	86.3
Corystus	M Eocene = Lutetian-Bartonian	47.8	37.8	Corystidae	Paleocene	66	23.03
Ctenocidaris	Danian	66	61.6	Ctenocidarinae	Paleocene	66	23.03
Cyclaster	Maastrichtian	72.1	66	Cyclasterinae	Maastrichtian	72.1	66
Dendraster	U Miocene = Tortonian-Messinian	11.63	5.33	Dendrasteridae	U Miocene = Tortonian-Messinian	11.63	5.33
Desorella	Oxfordian	163.5	157.3	Desorellidae	Oxfordian	163.5	157.3
Diadema	Recent	0	0	Diadematidae	Santonian	86.3	83.6
Diplocidaris	Pliensbachian	190.8	182.7	Diplocidaridae	Pliensbachian	190.8	182.7
Diplopodia	Aalenian	174.1	170.3	Diplopodiidae	Aalenian	174.1	170.3
Disaster	Callovian	166.1	163.5	Disasteridae	Callovian	166.1	163.5
Discoides	Aptian	126.3	113	Discoididae	Aptian	126.3	113
Echinarachnius	M Miocene = Langhian-Serravalian	15.97	11.63	Echinarachniidae	M Miocene = Langhian-Serravalian	15.97	11.63
Echinocardium	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Echinocardiidae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Echinocorys	Turonian	93.9	89.8	Echinocorythidae	Aptian	126.3	113
Echinocyamus	M Eocene = Lutetian-Bartonian	47.8	37.8	Echinocyamidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Echinolampas	M Eocene = Lutetian-Bartonian	47.8	37.8	Echinolampadidae	Maastrichtian	72.1	66
Echinometra	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Echinometridae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Echinoneus	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Echinoneidae	Paleocene	66	23.03
Echinus	U Miocene = Tortonian-Messinian	11.63	5.33	Echinidae	U Miocene = Tortonian-Messinian	11.63	5.33
Emiratia	Albian	113	100.5	Emiratiidae	Toarcian	182.7	174.1
Eodiadema	Sinemurian	199.3	190.8	Diadematidae	Santonian	86.3	83.6

Genus	First appearance of genus in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)	Family	First appearance of family in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)
Eoscutella	M Miocene = Langhian-Serravalian	15.97	11.63	Eoscutellidae	M Miocene = Langhian-Serravalian	15.97	11.63
Eotiaris	L Permian = Cisuralian	298.9	272.3	Miocidaridae	L Permian = Cisuralian	298.9	272.3
Eupatagus	L Eocene = Ypresian	56	47.8	Eupatagidae	L Eocene = Ypresian	56	47.8
Eurypatagus	Recent	0	0	Eurypataginae	Miocene	23.03	5.333
Faujasia	Campanian	83.6	72.1	Faujasiidae	Campanian	83.6	72.1
Fibularia	M Eocene =Lutetian-Bartonian	47.8	37.8	Fibulariidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Fossulaster	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Fossulasteridae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Galerites	Santonian	86.3	83.6	Galeritidae	Albian	113	100.5
Galeropygus	Toarcian	182.7	174.1	Galeropygidae	Toarcian	182.7	174.1
Gauthieria	Cenomanian	100.5	93.9	Phymosomatidae	Callovian	166.1	163.5
Glyphocyphus	L Aptian	126.3	122.98	Glyphocyphidae	Aptian	126.3	113
Glypticus	Callovian	166.1	163.5	Glypticidae	Toarcian	182.7	174.1
Glyptocidaris	Eocene	56	33.9	Glyptocidaridae	Eocene	56	33.9
Goniocidaris	Maastrichtian	72.1	66	Goniocidarinae	Maastrichtian	72.1	66
Goniophorus	U Albian	107.59	100.5	Hyposaleniinae	Oxfordian	163.5	157.3
Haimea	M Eocene = Lutetian-Bartonian	47.8	37.8	Oligopygidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Hemiaster	Albian	113	100.5	Hemiasteridae	Aptian	126.3	113
Hemicidaris	Toarcian	182.7	174.1	Hemicidaridae	Toarcian	182.7	174.1
Hemipneustes	Campanian	83.6	72.1	Hemipneustidae	Albian	113	100.5
Heterodiadema	Cenomanian	100.5	93.9	Heterodiadematidae	Bathonian	168.3	166.1
Histocidaris	Cenomanian	100.5	93.9	Histocidaridae	Cenomanian	100.5	93.9
Holaster	Hauterivian	133.9	130.8	Holasteridae	Hauterivian	133.9	130.8
Holectypus	Pliensbachian	190.8	182.7	Holectypidae	Pliensbachian	190.8	182.7
Holosalenia	Cenomanian	100.5	93.9	Holosaleniini	Cenomanian	100.5	93.9
Hyboclypus	Bajocian	170.3	168.3	[Hyboclypids]	Bajocian	170.3	168.3
Hygrosoma	Recent	0	0	Hygrosomatinae	Recent	0	0
Hyposalenia	Tithonian	152.1	145	Hyposaleniinae	Oxfordian	163.5	157.3
Kamptosoma	Recent	0	0	Kamptosomatidae	Recent	0	0
Laganum	Pliocene	5.333	2.588	Laganidae	Miocene	23.03	5.333
Lovenia	U Oligocene = Chattian	28.1	23	Loveniidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Macropneustes	M Eocene = Lutetian-Bartonian	47.8	37.8	Macropneustidae	L Eocene = Ypresian	56	47.8
Maretia	M Eocene = Lutetian-Bartonian	47.8	37.8	Maretiidae	M Eocene = Lutetian-Bartonian	47.8	37.8

Genus	First appearance of genus in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)	Family	First appearance of family in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)
Megapneustes	L Eocene = Ypresian	56	47.8	Macropneustidae	L Eocene = Ypresian	56	47.8
Mellita	U Miocene = Tortonian-Messinian	11.63	5.33	Mellitidae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Micraster	Turonian	93.9	89.8	Micrasteridae	Barremian	130.8	126.3
Micropyga	Recent	0	0	Micropygidae	L Eocene = Ypresian	56	47.8
Monophoraster	M Miocene = Langhian-Serravalian	15.97	11.63	Monophorasteridae	U Oligocene = Chattian	28.1	23
Neoglobator	Maastrichtian	72.1	66	Neoglobatoridae	Maastrichtian	72.1	66
Neolaganum	U Eocene = Priabonian	37.8	33.9	Neolaganidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Neolampas	Recent	0	0	Neolampadidae	L Eocene = Ypresian	56	47.8
Nucleolites	Bathonian	168.3	166.1	Nucleolitidae	Bajocian	170.3	168.3
Oligopygus	M Eocene = Lutetian-Bartonian	47.8	37.8	Oligopygidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Orthopsis	Bathonian	168.3	166.1	Orthopsidae	Bathonian	168.3	166.1
Ovulaster	Senonian	86.3	83.6	Ovulasteridae	Senonian	86.3	83.6
Palaeostoma	Eocene	56	33.9	Palaeostomatidae	L Eocene = Ypresian	56	47.8
Palaeotropus	Recent	0	0	Palaeotropidae	M Miocene = Langhian-Serravalian	15.97	11.63
Paleopneustes	Miocene	23.03	5.333	Paleopneustidae	Miocene	23.03	5.333
Paraphormosoma	Recent	0	0	Paraphormosomatinae	Recent	0	0
Parasalenia	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Parasaleniidae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Parechinus	Pleistocene	2.588	0.0118	Parechinidae	Miocene	23.03	5.333
Pelanodiadema	Bajocian	170.3	168.3	Pelanodiademinae	Bajocian	170.3	168.3
Periaster	Cenomanian	100.5	93.9	Periasteridae	Cenomanian	100.5	93.9
Pericosmus	L Eocene = Ypresian	56	47.8	Pericosmidae	L Eocene = Ypresian	56	47.8
Phormosoma	Recent	0	0	Phormosomidae	Recent	0	0
Phyllacanthus	U Oligocene = Chattian	28.1	23	Cidaridae	Bajocian	170.3	168.3
Phymosoma	Callovian	166.1	163.5	Phymosomatidae	Callovian	166.1	163.5
Plesiaster	Santonian	86.3	83.6	Micrasteridae	Barremian	130.8	126.3
Plesiolampas	U Paleocene = Thanetian	59.2	56	Plesiolampadidae	Maastrichtian	72.1	66
Plexechinus	Recent	0	0	Plexechinidae	Recent	0	0
Pliolampas	Eocene	56	33.9	[Pliolampadidae]	Maastrichtian	72.1	66
Polycidaris	Carnian	237	228.4	Polycidaridae	Carnian	237	228.4
Polydiadema	Toarcian	182.7	174.1	Emiratiidae	Toarcian	182.7	174.1
Poriocidaris	E Cenomanian	100.5	93.98	Histocidaridae	Cenomanian	100.5	93.9
Pourtalesia	M Miocene = Langhian-Serravalian	15.97	11.63	Pourtalesiidae	M Miocene = Langhian-Serravalian	15.97	11.63

Genus	First appearance of genus in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)	Family	First appearance of family in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)
Prenaster	L Eocene = Ypresian	56	47.8	Prenasteridae	Maastrichtian	72.1	66
Protoscutella	M Eocene = Lutetian-Bartonian	47.8	37.8	Protoscutellidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Pseudholaster	Valanginian	145	139.4	Pseudholasterinae	Berriasian	145	139.4
Pseudodiadema	Oxfordian	163.5	157.3	Pseudodiadematidae	Rhaetian	209.5	201.3
Pseudosalenia	Oxfordian	163.5	157.3	Pseudosaleniidae	Oxfordian	163.5	157.3
Pygaster	Bajocian	170.3	168.3	Pygasteridae	Bajocian	170.3	168.3
Pygaulus	Barremian	130.8	126.3	Pygaulidae	Valanginian	145	139.4
Pygorhytis	Toarcian	182.7	174.1	Pygorhytidae	Toarcian	182.7	174.1
Rhabdocidaris	Aalenian	174.1	170.3	Rhabdocidaridae	Aalenian	174.1	170.3
Roseicidaris	Toarcian	182.7	174.1	Psychocidaridae	Toarcian	182.7	174.1
Rotula	Recent	0	0	Rotulidae	Recent	0	0
Salenia	Barremian	130.8	126.3	Saleniidae	Barremian	130.8	126.3
Salenocidaris	L Paleocene = Danian	66	61.6	Salenocidarini	L Paleocene = Danian	66	61.6
Schizaster	L Eocene = Ypresian	56	47.8	Schizasteridae	Maastrichtian	72.1	66
Scutaster	M Eocene = Lutetian-Bartonian	47.8	37.8	Scutasteridae	M Eocene = Lutetian-Bartonian	47.8	37.8
Scutella	U Oligocene = Chattian	28.1	23	Scutellidae	U Eocene = Priabonian	37.8	33.9
Scutellina	M Eocene = Lutetian-Bartonian	47.8	37.8	Scutellines stem	M Eocene = Lutetian-Bartonian	47.8	37.8
Scutellinoides	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Scutellinoididae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Serpianotiaris	Anisian	247.1	241.5	Serpianotiaridae	Anisian	247.1	241.5
Somaliaster	Camp./Maast.	83.6	66	Somaliasteridae	Campanian	83.6	72.1
Spatangus	L Oligocene = Rupelian	33.9	28.1	Spatangidae	Eocene	56	33.9
Sperosoma	Recent	0	0	Sperosomatinae	Recent	0	0
Stegaster	Campanian	83.6	72.1	Stegasteridae	U Cretaceous	100.5	66
Stenonaster	U Cretaceous	100.5	66	Stenonasteridae	U Cretaceous	100.5	66
Stereocidaris	Aptian	126.3	113	Stereocidaridae	Aptian	126.3	113
Stigmatopygus	Campanian	83.6	72.1	Faujasiidae	Campanian	83.6	72.1
Stomechinus	Aalenian	174.1	170.3	Stomechinidae	Toarcian	182.7	174.1
Stomopneustes	Recent	0	0	Stomopneustidae	Bajocian	170.3	168.3
Strongylocentrotus	L Miocene = Aquitanian-Burdigalian	23.03	15.97	Strongylocentrotidae	L Miocene = Aquitanian-Burdigalian	23.03	15.97
Stylocidaris	Miocene	23.03	5.333	Cidaridae	Bajocian	170.3	168.3
Taiwanaster	Oligocene	33.9	23.03	Taiwanasteridae	Oligocene	33.9	23.03
Temnopleurus	Pliocene	5.333	2.588	Temnopleuridae	L Eocene = Ypresian	56	47.8
Tithonia	Bathonian	168.3	166.1	Tithoniidae	Bathonian	168.3	166.1
Toxaster	Valanginian	145	139.4	Toxasteridae	Valanginian	145	139.4

Genus	First appearance of genus in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)	Family	First appearance of family in fossil record (stage)	Lower Boundary (Ma)	Upper Boundary (Ma)
Toxopneustes	Pliocene	5.333	2.588	Toxopneustidae	U Oligocene = Chattian	28.1	23
Triadotiaris	Anisian	247.1	241.5	Triadotiaridae	Anisian	247.1	241.5
Trigonocidaris	Recent	0	0	Trigonocidaridae	U Eocene = Priabonian	37.8	33.9
Tylocidaris	U Albian	107.59	100.5	Psychocidaridae	Toarcian	182.7	174.1
Typocidaris	Aptian	126.3	113	Stereocidaridae	Aptian	126.3	113
Unifascia	M Eocene = Lutetian-Bartonian	47.8	37.8	Unifasciidae	M Eocene = Lutetian-Bartonian	47.8	37.8
Urechinus	Recent	0	0	Urechinidae	Recent	0	0
Zeuglopleurus	Cenomanian	100.5	93.9	Zeuglopleuridae	Albian	113	100.5

Table S2. Stage-based timescale where average time interval is about 10 million years. Ma = millions of years ago; my = millions of years. Lower and upper bounds based on Gradstein et al (21).

Stages	Lower Boundary (Ma)	Upper Boundary (Ma)	Duration (my)
Asselian-Sakmarian	298.8	291.1	7.7
Artinskian	291.1	279.3	11.8
Kungurian	279.3	272.3	7
Roadian-Capitanian	272.3	257.5	14.8
Wuchiapinian-Changhsingian	257.5	252.5	5
Induan-Olenekian	252.2	247.1	5.1
Anisian-Ladinian	247.1	237	10.1
Carnian	237	228.4	8.6
Norian	228.4	209.5	18.9
Rhaetian	209.5	201.3	8.2
Hettangian-Sinemurian	201.3	190.8	10.5
Pliensbachian	190.8	182.7	8.1
Toarcian	182.7	174.1	8.6
Aalenian-Callovian	174.1	163.5	10.6
Oxfordian-Kimmeridgian	163.5	152.1	11.4
Tithonian	152.1	145	7.1
Berriasian-Valanginian	145	133.9	11.1
Hauterivian-Barremian	133.9	126.3	7.6
Aptian	126.3	112.6	13.7
Albian	112.6	100.5	12.1
Cenomanian	100.5	93.9	6.6
Turonian-Santonian	93.9	83.6	10.3
Campanian	83.6	72.1	11.5
Maastrichtian	72.1	66	6.1
Danian-Thanetian	66	56	10
Ypresian	56	47.8	8.2
Lutetian-Bartonian	47.8	37.8	10
Priabonian-Rupelian	37.8	28.1	9.7
Chattian-Aquitanian	28.1	20.44	7.66
Burdigalian-Serravallian	20.44	11.63	8.81
Tortonian-present	11.63	0	11.63

Table S3. Computation of completeness. 1 = species within clade recorded from that time interval; 0 = no species within clade recorded from that time interval. Lower and upper boundaries of stages from Gradstein et al (21).

				Geologic stage (abbreviated)	Kun		Tar		Ind-Olen	Anis-Lad	Car	Nor	Rha	Hett-Sin	Plie	Toa	Aal-Cal	Oxf-Kim	Tith	Ber-Val	Haut-Bar	Apt	Alb	Cen	Tur-San	Ca m	Maa	Dan-Tha	Ypr	Lut-Bart	Pri-Rup	Char-Aqu	Bur-Ser	Tor-Plio
				Lower boundary of stage (Ma)	279	272	268	258	252	247	237	228	210	201	191	183	174	164	152	145	134	126	113	101	94	84	72	66	56	47.8	38	28.1	20	12
				Upper boundary of stage (Ma)	272	268	258	252	247	237	228	210	201	191	183	174	164	152	145	134	126	113	101	93.9	84	72	66	56	48	37.8	28	20.4	12	2.6
Node	**Description**	**Clades**	**Date**	**Oldest fossil**																														
Root	Cidaroid-euechinoid split	Cidaroid	255	Eocidaris keyserlingi	1		0		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
		Euechinoid			0		0		0	1	1																							
1	Echinothurioid-Acroechinoid split	Echinothurioid	220	Hemipedina' hudsoni Kier								0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1
		Acroechinoid																																
2	pedinoid-irreg/stiro split	Aulodont	210	Diadempsis serialis group								1	1	1	1	1	1	1	1	1	0	1	1	1	1	0	1	1	1	1	1	0	1	0
		Irregular+stiro										0	1																					
3	Diadematoid-irreg/stiro split	Pedinoid	210	Diadempsis serialis group										0																				
		Aspidodiadema												1																				
4	Diadematid-aspido split	Aspidodiadematids	210	Gymnotiara varusense group											1	0	1	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
		Diadematid													0	0	1	1	0	0	0	1	1	1	1	1	1	1	0	0	0	1	1	0
5	stirodont/irregular split	Stirodont	200	Jesionekechinus hawkinsi										1																				
		Irregulars												1																				
6	Echinoneoid-microstomata split	Echinoneoids	185	Galeropygus sublaevis (McCoy)													0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
		Microstomata																																
7	Atelostomate-neognath split	atelostomates	175	Hyboclypus													1	1	1	1														
		Neognaths															1	1	1	1	1	1												
8	Cassid-Clyp split	Cassid	105	Nucleopygus angustatus (Clark)																			1	1	1									
		Clypeasterids																					0	0	0									
9	Cassid-Echinolamp split	cassidulids	90	Hungaresia ovum (Grateloup)																					0	0	1	1	1	1	1	1	1	1
		echinolampadids																							1	1	1	0	1	1	1	1	1	1
10	Clyp-Scutelline split	cassid	105	Nucleopygus angustatus (Clark)																									1	1	1	1	1	1
		scutelline																																
11	Scutelline-Laganine split	scutellines	55	Eoscutum doncieuxi (Lambert)																														
		Laganines																																

| | | |

					Kun		Tar		Ind-Olen	Anis-Lad	Car	Nor	Rha	Hett-Sin	Plie	Toa	Aal-Cal	Oxf-Kim	Tith	Ber-Val	Haut-Bar	Apt	Alb	Cen	Tur-San	Cam	Maa	Dan-Tha	Ypr	Lut-Bart	Pri-Rup	Char-Aqu	Bur-Ser	Tor-Plio
				Geologic stage (abbreviated)																														
				Lower boundary of stage (Ma)	279	272	268	258	252	247	237	228	210	201	191	183	174	164	152	145	134	126	113	101	94	84	72	66	56	47.8	38	28.1	20	12
				Upper boundary of stage (Ma)	272	268	258	252	247	237	228	210	201	191	183	174	164	152	145	134	126	113	101	93.9	84	72	66	56	48	37.8	28	20.4	12	2.6
Node	**Description**	**Clades**	**Date**	**Oldest fossil**																														
25	Temno-Toxo split	temnopleurid	100	Zeuglopleurus costulatus Gregory																							1	1	1	1	1	1	1
		lytechinus																									0	0	1	1			
26	Genocid-Cytechin split	genocidaris	20	Arbacina monilis (Desmarest)																											1	1	1
		cyrtechinus																													0	0	0
				Number of lineages captured by record	1		0		1	2	2	2	3	5	5	2	9	10	5	9	6	9	9	10	11	8	11	12	13	19	18	20	23	23
				Phylogenetic estimate of lineages present	2		2		2	2	2	4	4	6	8	8	11	11	11	12	12	12	13	15	16	16	16	17	20	25	25	29	29	29
				% complete (lineages captured by record)	0.5		0		0.5	1	1	0.5	0.75	0.83	0.63	0.25	0.82	0.91	0.45	0.75	0.5	0.75	0.69	0.67	0.7	0.5	0.7	0.7	0.7	0.76	0.7	0.69	0.8	0.8

Table S4. Numbers of collections yielding specific echinoid clades. Tabulation based on records downloaded from the Paleobiology Database (www.paleobiodb.org) on 12 November 2014. Lower and upper age boundaries based on Gradstein et al (21).

Stages	Lower Boundary (Ma)	Upper Boundary (Ma)	Cidaroids	Regular carinaceans	regular echinoids	Irregularia	Any echinoids
Induan-Olenekian	252.2	247.1	35	0	35	0	35
Anisian-Ladinian	247.1	237	36	0	37	0	37
Carnian	237	228.4	27	0	27	0	27
Norian	228.4	209.5	7	0	8	0	8
Rhaetian	209.5	201.3	4	0	4	0	4
Hettangian-Sinemurian	201.3	190.8	3	0	15	1	16
Pliensbachian	190.8	182.7	4	1	6	0	6
Toarcian	182.7	174.1	1	2	7	0	7
Aalenian-Callovian	174.1	163.5	43	104	125	83	162
Oxfordian-Kimmeridgian	163.5	152.1	72	99	148	83	197
Tithonian	152.1	145	5	10	14	5	18
Berriasian-Valanginian	145	133.9	15	15	23	40	46
Hauterivian-Barremian	133.9	126.3	32	55	67	158	184
Aptian	126.3	112.6	23	34	41	75	91
Albian	112.6	100.5	17	29	45	100	121
Cenomanian	100.5	93.9	29	54	60	118	128
Turonian-Santonian	93.9	83.6	24	41	60	261	287
Campanian	83.6	72.1	13	24	28	91	102
Maastrichtian	72.1	66	22	51	60	164	180
Danian-Thanetian	66	56	20	20	31	59	72
Ypresian	56	47.8	13	4	16	22	30
Lutetian-Bartonian	47.8	37.8	32	20	43	140	149
Priabonian-Rupelian	37.8	28.1	17	12	28	153	163
Chattian-Aquitanian	28.1	20.44	9	5	13	54	60
Burdigalian-Serravallian	20.44	11.63	20	14	29	114	121
Tortonian-present	11.63	0	23	63	80	180	228

Table S5. Correlations between rates of character change and completeness (upper panel) and rates of character change and sampling intensity (lower panel). Shorthand for scaling methods is followed by panel label in Figure S3.

Scaling method	Spearman's Rho	p-value
equal (E)	-0.238	0.252
prop (F)	-0.206	0.324
aba (B)	0.363	0.074
mbl (D)	-0.062	0.767
zba (C)	0.281	0.174
stand (A)	0.006	0.977
cal3-noA (G)	0.059	0.780
cal3-withA (H)	-0.058	0.784
equal (E)	-0.294	0.154
prop (F)	-0.105	0.617
aba (B)	0.244	0.239
mbl (D)	-0.067	0.750
zba (C)	0.035	0.867
stand (A)	-0.126	0.546
cal3-noA (G)	-0.128	0.539
cal3-withA (H)	-0.136	0.515

Table S6. Tabulation of character changes during the early evolution of clypeasteroids. Below are listed all characters associated with the first 7 dichotomies at the base of the clypeasteroids (from the oligopygoid-clypeasteroid divergence to the establishment of the three major subclades, Clypeasterina, Laganiformes, and Scutteliformes). There are 51 characters in all. We scored each as (i) feeding related (e.g., peristome shape, tube-foot arrangement, lantern or perignathic girdle features); (ii) hydrodynamic features (i.e. to do with life in high energy environments, e.g., test flattening, internal butressing, lunules); (iii) neutral with respect to (i) and (ii) (e.g., detailed plating differences that have no clear functional correlate). Characters A27 and A28 are conservatively scored as neutral although it is possible that they are related to cilicary current generation during feeding.

Character		Feeding	Hydro	Neutral
A2	Corona dorso-ventrally flattened:		X	
A7	Lunules	X	X	
A27	Glassy tubercles developed on the aboral surface			?
A28	Glassy tubercles developed on the oral surface			?
A30	Internal radial bars at adradial edge of interambulacral		X	
A31	Internal buttresses in ambulacral zones		X	
A32	Internal buttresses in interambulacral zones		X	
A33	Lantern and/or pharynx and gut separated by internal pillars		X	
B18	Shape of periproct			X
B24	Gonopores opening fully within genital plate			X
B30	Hydropores: opening in groove			X
B44	Periproct position in adult	X		
B45	Position of base of periproct in adult	X		
C2	Ambulacral plating: successive elements within zone similar			X
C8	Number of elements involved in compound plate			X
C9	Maximum number of demiplates or occluded plates in a compound:			X
C15	succeeding plates	X		
C16	Ambulacral width at ambitus	X		
C18	First pair of ambulacral plates in basicoronal ring paired:			X
C21	Ambulacral projection into peristome at perradius	X		
C22	Presence of superficial amb plates down adradius	X		
C34	More than one tube foot and microscopic pore present on ambulacral plates at ambitus	X		
C35	Multiple microscopic tube feet/pores present within the adapical region	X		
C37	Pores and tube-feet fields extend beyond ambulacral zones and onto interambulacral plates	X		
C41	Three pores passing through first ambulacral plates:			X
C47	Skeletal supports at tip of suckered tube-feet	X		
D4	Interambulacra 1 and 4 on oral surface: plate columns contiguous			X
D5	Interambulacra 2 and 3 plating on oral surface: contiguous			X

Character		Feeding	Hydro	Neutral
D6	Interambulacra 5 plating on oral surface: contiguous			X
D13	In interambulacrum 5, plates 2a and 2b separated by: an oblique suture			X
D14	In interambulacrum 5, plates 2a and 2b paired and of similar size			X
D15	In interambulacrum 5, basicoronal (labrum) plate shape			X
D17	In interambulacrum 5, episternal plates paired and opposite			X
D22	Interambulacra end adapically in a single plate			X
D23	Posterior suture of interambulacral plate 4.b.2 lies aborally of the posterior suture of plate 4.a.2			X
D24	Posterior suture of interambulacral plate 1.b.2 lies aborally of the posterior suture of plate 1.a.2			X
D29	Number of ambulacral plate abutting rear suture of plate 5.b.2 in ambulacrum I:			X
D30	Interambulacra with a zone of much narrower plates at the ambitus			X
E2	Peristomal margin in interambulacrum 5 inturned	X		
E3	Peristome outline	X		
E5	Presence of swollen interambulacra indenting all five faces of the peristome	X		
E13	Five valve-like plates without associated tube-feet cover the peristome			X
E17	Perignathic girdle includes apophyses developed from the pair of perioral interambulacral plates	X		
E18	Apophyses: separate adradial flanges widely separated and not connected interradially	X		
F3	Lantern in plan view	X	X	
F14	Hemipyramids with internal wing-like extensions buttressed by flanges	X		
G10	Secondary spines: end in blunt rounded point without gland	X		
G17	Spines with swollen and bent head present:	X		
G20	Sphaeridia: superficial on plate surface			X
G21	Sphaeridia: one per ambulacrum			X
G23	Sphaeridia position: adradial (beside pore-pairs)			X

Table S7. Stage-based timescale where average time interval is approximately 5 million years. Ma = millions of years ago; my = millions of years. Lower and upper age boundaries based on Gradstein et al (21).

Geologic stage or epoch	Lower boundary (Ma)	Upper boundary (Ma)	Duration (my)
Asselian-Sakmarian	298.9	295.5	3.4
Sakmarian	295.5	290.1	5.4
Artinskian	290.1	279.3	10.8
Kungurian	279.3	272.3	7
Roadian	272.3	268.2	4.1
Wordian-Capitanian	268.2	257.5	10.7
Lopingian	257.5	252.2	5.3
Induan-Olenekian	252.2	247.1	5.1
Anisian	247.1	241.5	5.6
Ladinian	241.5	237	4.5
Carnian	237	228.4	8.6
Norian	228.4	209.5	18.9
Rhaetian	209.5	201.3	8.2
Hetttangian	201.3	199.3	2
Sinemurian	199.3	190.8	8.5
Pliensbachian	190.8	182.7	8.1
Toarcian	182.7	174.1	8.6
Aalenian-Bajocian	174.1	168.3	5.8
Bathonian-Callovian	168.3	163.5	4.8
Oxfordian	163.5	157.3	6.2
Kimmeridgian	157.3	152.1	5.2
Tithonian	152.1	145	7.1
Berriasian	145	139.4	5.6
Valanginian	139.4	133.9	5.5
Hauterivian	133.9	130.8	3.1
Barremian	130.8	126.3	4.5
Aptian	126.3	112.6	13.7
Albian	112.6	100.5	12.1
Cenomanian	100.5	93.9	6.6
Turonian	93.9	89.8	4.1
Coniacian	89.8	85.8	4
Santonian	86.3	83.6	2.7
Campanian	83.6	72.1	11.5
Maastrichtian	72.1	66	6.1
Danian	66	61.6	4.4
Selandian	61.6	59.2	2.4
Thanetian	59.2	56	3.2
Ypresian	56	47.8	8.2
Lutetian	47.8	41.2	6.6
Bartonian	41.2	37.8	3.4
Priabonian	37.8	33.9	3.9
Rupelian	33.9	28.1	5.8
Chattian	28.1	23	5.1
Aquitanian	23	20.44	2.56
Burdigalian	20.44	15.97	4.47
Langhian	15.97	13.82	2.15
Serravallian	13.82	11.63	2.19
Tortonian	11.63	7.25	4.38
Messinian	7.25	5.333	1.917
Pliocene	5.333	2.588	2.745
Pleistocene	2.588	0	2.588